

15. INSERTING THE MISSING CHARACTER

In such type of questions, a figure, a set of figures or a matrix is given, each of which bears certain characters, be it numbers, letters or a group of letters/numbers, following a certain pattern. The candidate is required to decipher this pattern and accordingly find the missing character in the figure.

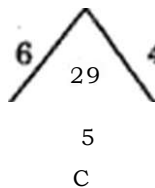
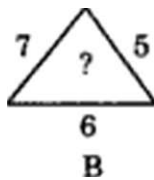
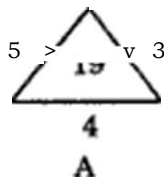
ILLUSTRATIVE EXAMPLES

Directions : Find the missing character from among the given (dternatives).
Ex. 1.



(a) 625 (6) 25 (c) 125 (d) 156
Sol. Clearly, $(3 + 2)^2 = 25$; $(15 + 6)^2 = (21)^2 = 441$; $(10 + 7)^2 = (17)^2 = 289$.
So, missing number = $(12 + 13)^2 = (25)^2 = 625$.
Hence, the answer is (a).

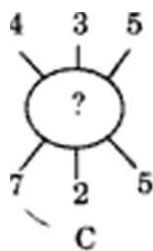
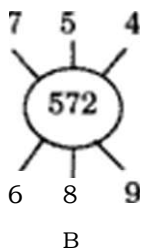
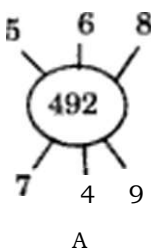
Ex. 2.



(a) 25 (b) 37 (c) 41 (d) 47
(Assistant Grade, 1995)

Sol. Clearly, in fig. (A), $5 \times 3 + 4 = 19$.
In fig. (C) = $6 \times 4 + 5 = 29$.
In fig. (B), missing number = $7 \times 5 + 6 = 35 + 6 = 41$.
Hence, the answer is (c).

Ex. 3.



(a) 115 (6) 130 (c) 135 (d) 140
(8.S.C. 1994)

Sol. Clearly, the number inside the circle is equal to the sum of the product of the upper three numbers and the product of the lower three numbers. Thus.

Then, $(1^2 + i^2) - (5^2 + x^2) = -12$
or $17 + 12 = (5^2 + x^2)$ or $x^2 = 29 - 25 = 4$ or $x = 2$.
Hence, the answer is (c).

Ex. 7.

3C	2B	4A
27A	?	64B
9C	4A	16B

- (a) 8C
- (b) 12B
- (c) 16C
- (d) 18C

Sol. In each row, out of the letters A, B and C, each of these must appear once. Also, in each column, the product of first and third numbers is equal to the second number. So, the missing number will be (2×4) i.e. 8 and the letter will be C. Thus, the answer is 8C.
Hence, the correct answer is (a).

EXERCISE 15

Directions (Questions 1 to 10) : Find the missing character in each of the following questions :

1.



(a) 4
(c) 343

(b) 305
(d) 729

(Asstt. Grade, 1994)

2.



(a) 72
(c) 68

(b) 70
(d) 66

(C.B.I. 1995)

3.



(a) 10
(c) 12

(b) 11
(d) 13

(P.C.S. 1995)

4.

x	c		
1	f	144	49 4
2		?	36 / 5

(a) 100
(c) 64

(b) 81
(d) 121

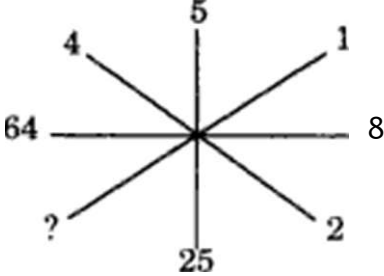
15.

5 / 1	7 s
64 v 3 \	
\ 11 27	? / 8 J
	2

(a) 0
(c) 125

(b) 8
(d)

6.

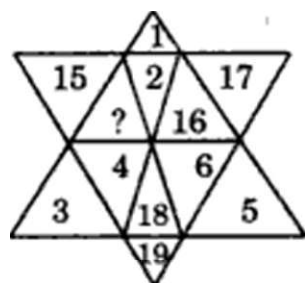


(a) 1
(c) 3

(b) 2
(d) 4

(C.B.I. 1997)

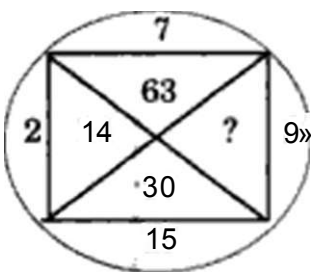
7.



- (a) 13 (b) 14
(c) 20 (d) 21

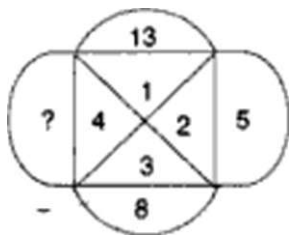
(CJLT. 1997)

8.



- (a) 33 (b) 145
(c) 135 (d) 18

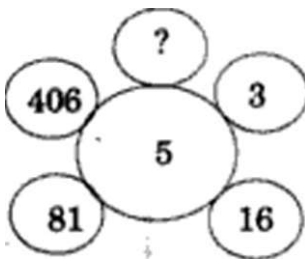
9.



- (a) 10 (b) 11
(c) 12 (d) 13

(Asstt. Grade. 1993), ,w

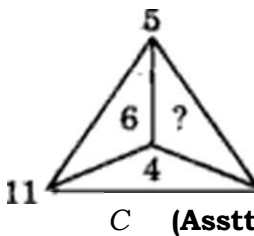
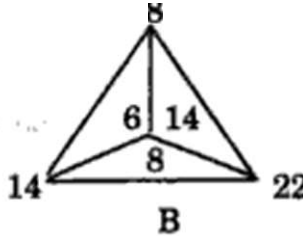
10.



- (a) 1 (b) 731
(c) 1625 (d) 2031

Directions (Questions 11 to 34) : In each of the following questions, a set of figures carrying certain characters, is given. Assuming that the characters in each set follow a similar pattern, find the missing character in each case.

11.

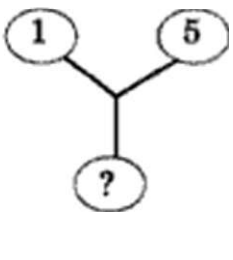
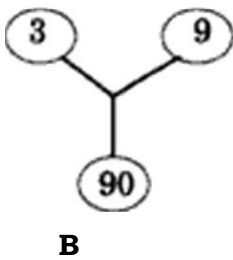
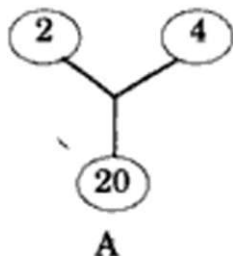


- (a) 6 (b) 8

- (c) 10

(Asstt. Grade, 1995)
(</) 14

12.



- (a) 75

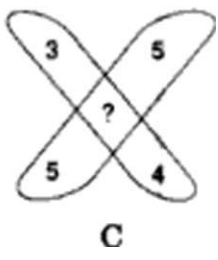
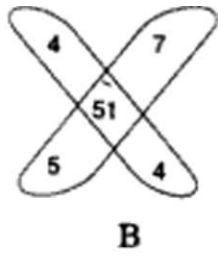
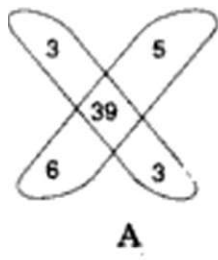
- (b) 26

- (c) 25

- (d) 20

Transmission Executives' 1994

13.



- (a) 47

- (b) 45

- (c) 37

- (</) 35

(C.B.I. 1995)

14.

915

551

3^4

A

543

246

789

B

863

?

241

C

(Asstt. Grade, 1997)

(a) 262

(b) 622

(c) 631

(d) 824

15.

56

12

4

A

67

21

5

B

48

?

10

C

(I. Tax A Central Excise, 1994)

(a) 14

(b) 22

(c) 32

(d) 320

16.

15

2

80

5

6

9

7

65

4

6

13

16

11

8

B

(a) 48

(b) 72

(c) 35

(d) 120

17.

128

80

A

167

207

B

12521

?

C

(S.S.C. 1994)

(a) 184

(b) 210

(c) 241

(d) 425

18.

6

A

B

?

C

(C.B.I. 1996)

(a) 18

(b) 12

(c) 9

(d) 6

19.

1

5

20

9

24

11

?

13

f

3

8

y

h

(Asstt. Grade, 1998)

(a) 117

(b) 36

(c) 32

(d) 26

20.

165

131

234

A

220

248

468

B

533

?

791

C

(9.8.C. 1996)

(a) 320

(b) 274

(c) 262

(d) 132

21.

36

^ 49

26

64

25

A

81

21

25

16

B

25

64

144

36

C

(M.B.A. 1998)

- (a) 19
- Cb) 23
- (c) 25
- (d) 31

22.

2

5

4

584

2

1

2

A

2

3

3

694

3

2

2

B

4

6

7

?

1

x

2

C

- (a) 937
- (6) 824
- (c) 769
- (d) 678

23.

2

3 ^ 3 3 ^ 2

4

A

©

5

B

VI

O

- (a) 78
- (6) 82
- (c) 94
- (d) 86

24.

f 251 17 \

A

18 \

8

B

/ 8 9 16

v / f

c

(a) 13

(6) 15

(c) 17

<d) 19

25.

v 43

38

3 5 \ /

48 .

?

^ S

6 6 ^

B

- (a) 127
- (6) 142
- (c) 158
- (d) 198

26.

fas

27

63

3

A

79

38

37

4

B

67

16

42

?

C

(C.B.I. 1995)

- (a) 5
- (6) 6
- (c) 8
- (d) 9

(Asstt. Grade, 1997)

28.

(a)

84	
14	12

(b) 100

81	
18	9

(c) 78

88	
?	11

(d) 63

(S.C.R.A. 1996)

C

29.

(a)

(b) 21

(c) 61

(d) 81

0

2

2

C

(L Tax. 1994)

30.

(a)

29	27	
39	80	33
45		43

(b) 70

29	30	
42	70	31
43		44

(c) 11

59	40	
?	80	10
39		20

(d) 12

(C.B.I. 1994)

C

31.

(a)

4	9
6	

(b) 50

9	16
12	

(c) 60

16	?
20	

(d) 69

(Transmission Executives', 1994)

32.

<a>

	5	
6	93	15
	3	

(b) 50

	7	
9	?	5
	6	

(c) 25

	18	
4	50	1
	8	

<d> 21

A

B

33.

(a) 5

	5	
16	109	2
	6	

(b) 19

	21	
22	53	19
	15	

(c) 27

	51	
17	?	48
	13	

(d) 89

(a) 7

25

49

129

34.

4 - (C a s V 3

5

A

4

B

(a) 2

(b) 3

(c) 4

(d) 5

Directions (Questions 35 to 74): In each of the following questions, a matrix of certain characters is given. These characters follow a certain trendy row-wise or column-wise. Find out this trend and choose the missing character accordingly.

35.

18	24	32
12	14	16
3	?	4
72	112	128

(a) 2

(b) 3

(c) 4

(d) 5

(U.D.C. 1995)

37.

28	60	48
5	6	7
14	39	27
7	?	16

(a) 18

(b) 23

(c) 24

(d) 27

(I. Tax & Central Excise. 1996)

39.

5	6	7
3	4	5
9	10	11
345	460	?

(a) 535

(b) 577

(c) 755

(d) 775

(Asstt. Grade, 1997)

41.

13	12	5
17	15	8
25	24	?
29	21	20

(a) 7

(b) 9

(c) 11

(d) 15

36.

3	6	8
5	8	4
4	7	?

(a) 6

(b) 7

(c) 8

(d) 9

(Asstt Grade, 1995)

38.

4	5	6
2	3	7
1	8	3
21	98	?

(a) 94

(c) 73

(d) 16

(C.B.I. 1995)

40.

6	6	8
5	7	5
4	3	?
120	126	320

(a) 4

(b) 8

(c) 12

(d) 16

(CJ3.1.1994)

42.

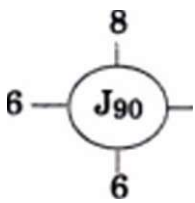
6	9	15
8	12	20
4	6	?

(a) 5

(b) 10

(e) 15

(d) 21



2	4	0
1	2	4
3	1	3
36	?	91

- (a) 25

(c) 59
- v

(rf) 73
- (I. Tax & Central Excise, 1995)

6	11	25
8	6	16
12	5	?

- (a) 18

(c) 12
- (6) 16

id) 10
- (Asstt. Grade, 1998)

1	3	7
5	12	14
25	<i>f</i>	28
125	192	56

- (a) 64

(c) 48
- (b) 56

(d) 40
- (Asstt. Grade, 1996)

13	54	?
7	45	32
27	144	68

- (a) 42

(c) 6
- (6) 36

(d) 4
- (S.S.C. 1996)

7	4	5
8	7	6
3	3	<i>f</i>
29	19	31

- (a) 3

(c) 5
- (b)* 4

(d) 6
- (U.D.C. 1995)

48.

3	15	4
7	38	5
3	?	5

- (a) 15

(c) 19
- (b)* 18

(d) 20
- (Asstt. Grade, 1995)

42	44	38
23	55	28
37	?	39

- (a) 22

(c) 66
- (b)* 33

(d) 77
- (I. Tax & Central Excise, 1996)

50.

1	7	9
2	14	<i>t</i>
3	105	117

- (a)* 26

(c) 16
- (b)* 20

(d) 12
- (Asstt. Grade, 1997)

963	2	844
464	?	903

- (a) 1

(c) 3
- (6) 2

(d) 4
- (I. Tax & Central Excise, 1994)

1	2	3
11	7	5
120	45	?

- (a) 19

(c) 16
- (6) 17

(</) 15
- (Asstt. Grade, 1994)

53.

51	11	61
64	30	32
35	?	43

- (a) 25
- (b) 27
- (c) 32
- (d) 37

55.

28	20	7
84	35	12
45	?	9

- (a) 15
- (b) 18
- (c) 20
- (d) 25

57.

12	(47)	21
10	(52)	4
64	(?)	24

- (a) 16
- (b) 40
- (c) 62
- (d) 83

(C.B.I. 1993)

59.

3	4	5
3	7	12
3	?	22

(Asstt. Grade, 1994)

- (a) 11
- (b) 10
- (c) 9
- (d) 8

61.

1	4	9	•> A
1	2	3	4
2	4	6	?

- (a) 16. 8
- (b) 49, 7
- (c) 36. 4
- (d) 25, 5

(Asstt. Grade, 1995)

54.

72	24	6
96	16	12
108	?	18

- (a) 12
- (b) 16
- (c) 18
- (d) 20

56.

?	13	49
9	17	69
13	11	59

- (a) 5
- (b) 9
- (c) 10
- (d) 21

58.

11	6	8
17	12	.
25	34	19
19	28	11

- (a) 9
- (b) 13
- (c) 15
- (d) 16

(Asstt. Grade, 1997)

60.

3	2	2
6	20	4
12	25	64
6	10	?

- (a) 6
- (b) 8
- (c) 12
- (d) 10

62.

7	9	21	27
4	2	36	18
9	4	54	?

- (a) 18
- (b) 24
- (c) 36
- (d) 58

(Tax & Central Excise, 1995)

A	D	H
F	I	M
9	N	R

- ia) K
- (6)N
- ic) 0
- id) P

F	.1	0
A	J	K

E M ? i

- ia) P
- ib) R
- ic) S
- id) V

B	G	N
D	J	R
G	N	4

- (a) U
- ib)V
- ic) W
- < d) X

(Asstt. Grade, 1996)

7B	5C	6B
3C	9B	19A
15 A	17A	?

- (a) IOC
- ib) 12C
- (c) 14B
- (d) 16C

Z4	X3	V9
A6	C2	?
T5	R4	P15

- (a) E10
- (6) E12
- (c) S10
- id) S12

BD3	CES	DFIS
EG ₂	FH ₄	GIS
HJ ₄	IKS	9

- ia) JL24
- ib) IJ18
- (c) JKis
- id) JL12

64.	Z	?	S
	R	0	9
	?	G	c

- (a) WJK
- (6) KWT
- (c) WKJ
- id) **JKW (C.B.I. 1995)**

66.	H	K	Q
	C	G	O
	E	J	9 i

- (a)T
- (6) P
- (c)N
- (d) L

68.	P	" T	?
	0	Q	s
	M	N	R

- ia) I
- ib) L
- (0 0
- id) U

70.	4C	2B	3A
	28A	?	45B
	7C	5A	15B

- (a) IOC
- (6) 12C
- (c) 13C
- id) 7C

72.	3C	27D	9E
	71	21K	3M
	4D	?	7J

- (a) 11E
- (6) 28G
- ic) 351
- (d) **48F (C.B.I. 1997)**

74. In the matrix given below, the values of A,B and C are

9	A	12
B	10	7
8	C	11

(Hotel Management, 1992)

- (o) A = 13, B = 11, C = 9

(c) A = 9, B = 11, C = 13
- (6) A = 13, B = 9, C = 11

(d) A = 9, B = 13, C = 11

Directions (Questions 75 to 78): In the following questions, numbers have been arranged according to the same general pattern. Find the missing number in each question.

75.

	310	
296		81
	215	14

(o) 6 (6) 7

(c) 9 (d) 18
76.

	72	
		16
	48	8

(a) 32 (6) 40

(c) 64 (d) 80
77.

	225	144	
12	81	15	

(o) 75 (6) 240

(c) 360 (d) 400
78.

	132	
	121	49
		11

(a) 5 (6) 15

(c) 18 (d) 72

Directions (Questions 79 to 81) : In each of the following questions, the numbers have been arranged according to the pattern shown in the sample figure given below. Find the missing number.

	104	
8	85	64
	13	

Sample Figure

79.

	154	
14	221	196
	?	

(a) 11 (6) 13

(c) 15 (rf) 17
80.

	315	
15	261	?
	21	

(a) 125 (6) 90

-(c) 105 (d) 225
81.

	9	
3	39	9
	27	

(a) 33 (6) 81

(c) 243 (d) 42

Directions (Questions 82-83): Study each of the following tables and choose the alternative which can best replace the sign of interrogation (?)

82.

3	8	10	2	?	1
6	56	90	2	20	0

(Railways, 1993)

(a) 0

(6) 3

(c) 5

(d) 7

83.

1	2	3	2	10	12
2	5	12	10	16	13
1	2	1	?	10	24

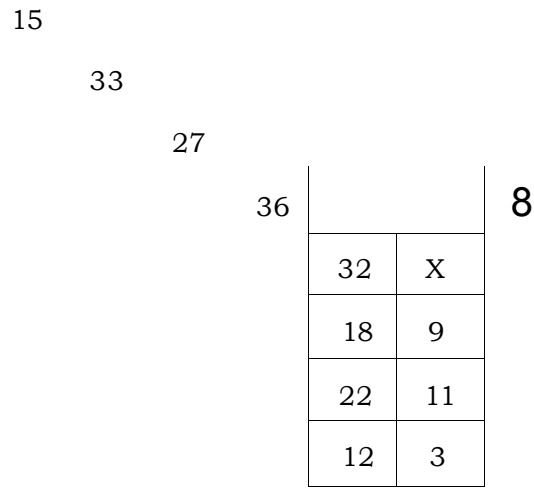
(a) 5

<b) 11

(c) 13

(d) 8

84. Find the value of X in the following figure :



- (a) 3
- (6) 4
- (c) 8
- (d) 12

ANSWERS

1. (c) : Moving clockwise, we get the sequence : $2^3, 3^3, 4^3, 5^3, 6^3, 7^3$
2. (6): Clearly, we have : $7 \times 2 \cdot 2 \gg 16$; $16 \times 2 \cdot 2 = 34$ and so on.
So, missing number - $34 \times 2 \cdot 2 = 70$.
3. (6): The numbers in the right half form the series : 2, 3, 4, 5.
The numbers in the left half form the series : 5, 7, 9, 11.
4. (a): Clearly. $(7 + 5)^2 = 144$; $(3 + 4)^2 = 49$; $(5 + 1)^2 = 36$.
So, missing number = $(2 + 8)^2 = 100$.
5. id): Clearly. $(5 - 4)^3 = 1$; $(7 - 3)^3 = 64$; $<11 - 8)^3 = 27$.
So, missing number « $(8 - 2)^3 \cdot 6^3 \cdot 216$.
6. (a): The two ends of each line segment contain a number and its square.
So, missing number = $1^2 = 1$.
7. ib): The given figure contains numbers 1 to 6 in three alternate segments, the smaller number being towards the outside and the numbers 14 to 19 in the remaining three alternate segments with the smulle- number towards the inside.
8. (c): Clearly, we have : $15 \times 2 = 30$. $2 \times 7 = 14$, $7 \times 9 = 63$.
So, missing number = $9 \times 15 = 135$.

9. (c): The arrangement is $5 \cdot 3 = 8$, $8 \cdot 4 = 12$, $12 \cdot 1 = 13$.
So, the missing number is 12.
10. Id): Clearly, we have : $(3 \times 5) + 1 = 16$, $(16 \times 5) + 1 = 81$, $(81 \times 6) + 1 = 406$
So, missing number = $(406 \times 5) + 1 = 2031$.
11. le): In fig. <A>. $10 - 4 = 6$, $18 - 10 = 8$, $18 - 4 = 14$.
In fig. . $14 - 8 = 6$, $22 - 14 = 8$, $22 - 8 = 14$.
In fig. (C). $11 - 5 = 6$, $15 - 11 = 4$.
So, missing number = $15 - 5 = 10$.
12. (b): The lower number is obtained by adding the squares of the upper two numbers. Thus,
In fig. (A). $2^2 + 4^2 = 20$.
In fig. (B). $3^2 + 9^2 = 90$
In fig. (C). missing number = $1^2 + 6^2 = 37$.
13. (c): In fig. (A), $(3 \times 3) + (6 \times 5) = 39$.
In fig. (B). $(4 \times 4) + (6 \times 7) = 51$.
In fig. (C). missing number = $(3 \times 4) + (5 \times 5) = 37$.
14. (b): In fig. (A). $(915 - 364) = 551$.
In fig. (B). $(789 - 543) = 246$.
In fig. (C), missing number = $(863 - 241) = 622$.
15. (c) : The number inside the triangle is obtained by dividing the product of the numbers along the sides of the triangle by 10. Thus,
In fig. (A). $(5 \times 6 \times 4) \div 10 = 12$.
In fig. (B), $(6 \times 7 \times 5) \div 10 = 21$.
In fig. (C). missing number = $(4 \times 8 \times 10) \div 10 = 32$.
16. (a): In fig. (A). $(15 - 5) \times (2 - 6) = 80$.
In fig. (B). $(9 - 4) \times (7 - 6) = 5$.
In fig. (C). missing number = $(13 - 11) \times (16 - 8) = 48$.
17. (a): The lower number is the difference of the squares of the upper two numbers. Thus.
In fig. <A>. $12^2 - 8^2 = 80$.
In fig. (B). $16^2 - 7^2 = 207$.
In fig. (C). missing number = $25^2 - 21^2 = 625 - 441 = 184$.
18. te): The above three numbers are multiples of the number at the bottom.
Clearly, 36, 18 and 27 are all multiples of 9
So, the missing number is 9.
19. (i) : In fig. (A). $5 \times 4 = 20$, $5 \cdot 4 = 9$.
In fig. (B). $3 \times 8 = 24$, $3 \cdot 8 = 11$.
In fig. (C). missing number = $9 \times 4 = 36$.
20. (c): The digits of the number inside the circle are the differences between the corresponding numbers above and below the circle. Thus.
In fig. (A). $1 = (2 - 1) - (3 - 6) = 5 - 4$.
In fig. (B). $2 = (4 - 2) - (6 - 8) = 2 - 0$.
So, in fig. (C). the digits of the missing number are :
 $(7 - 5) = 2$, $(9 - 3) = 6$, $(3 - 1) = 2$.
Missing number is 262.
21. Id): In fig. (A). $6^2 \times 36 = 1296$, $8^2 = 64$, $6^2 = 36$, $7^2 = 49$ And, $6 + 8 + 5 + 7 = 26$.
In fig. (B). $3^2 = 9$, $5^2 = 25$, $4^2 = 16$, $9^2 = 81$ And, $3 + 5 + 4 + 9 = 21$,
In fig. (C). $5^2 = 25$, $12^2 = 144$, $6^2 = 36$, $8^2 = 64$.
So, missing number = $5 \times 12 \times 6 \times 8 = 2880$.

22. (d): In fig. (A). $5 * 1 \gg 5$, $4 * 2 = 8$, $2 - 2 = 4$.
In fig. (B), $3 * 2 = 6$, $3 \times 3 = 9$. $2 * 2 = 4$.
/. In fig. (C), the digits of the missing number are $6 \times 1 = 6$. $7 \times 1 = 7$. $4 * 2 = 8$.
So, missing number - 678.
23. (d): The sum of square* of numbers outside the circle is equal to the number inside the circle.
In fig. (A), $2^2 + 2^2 \cdot 4^2 \cdot 3^2 \cdot 4 \cdot 4 \cdot 16 \cdot 9 = 33$.
In fig. (B). $3^2 \cdot 2^2 + 5^2 \cdot 4^2 = 9 + 4 \cdot 25 + 16 = 54$
In fig. (C), missing number $\cdot 6^2 \cdot 5^2 \cdot 4^2 \cdot 3^2 \cdot 36 \cdot 25 \cdot 4 \cdot 16 \cdot 9 = 86$.
24. (6): The sum of the numbers in the upper two parts of each circle is seven times the number in the third part.
In fig. (A), $(25 \cdot 17) \ll 42 - 7 \times (6)$.
In fig. (B). $(38 \cdot 18) \gg 56 = 7 \times (8)$.
In fig. (C). $(89 \wedge 16) = 105 = 7 \times (15)$.
∴ The missing number is 15.
25. (6): In fig. (A). $(101 \ 4 \ 15) - (35 \cdot 43) \ll 116 - 78 \gg 38$.
The game pattern would be followed in fig. (B).
Missing number - $(48 \cdot 184) - (56 + 34) = 232 - 90 = 142$.
26. id): In fig. (A), $93 - (27 \cdot 63) = 3$.
In fig. (B). $79 - (38 + 37) = 4$.
In fig. (C), missing number $\gg 67 - (16 \cdot 42) \otimes 9$
27. (c>): In fig. (A). $<9 \times 10) - (4 * 8) \gg 58$.
In fig. (B). missing number $\cdot (15 \times 10) - (9 \times 8) \cdot 150 \quad 72 - 78$
28. (a): In fig. (A). $1 \ 2 \times \wedge - 8 \ 4$

In fig. (B). $9k \ y = 81$.

Let the missing number in fig. (C) be x.

Then. $11x \wedge .88 \text{ or } x \ll -J - - - 16$.
29. (c): The number inside the circle is equal to the difference between the sum of the numbers at the extremities of the horizontal diameter and the sum of numbers at the extremities of the vertical diameter.
In fig. (A). $(5 \wedge 6) - (7 \cdot 4) \ll 0$.
In fig. (B). $(7 + 6) - (8 + 4) \gg 1$.
In fig. (C), missing number $= (11 \cdot 2) - (0 + 2) = 11$.
30. (d): The sum of numbers at the extremities of the three line segments in each figure is same.
Thus, in fig. (A), $39 \cdot 33 * 29 * 43 = 27 - 45$.
In fig. (B). $42 + 31 = 29 * 44 = 30 + 43$.
Let the missing number in fig. (C) be x.
Then, $x \cdot 10 = 59 * 20 \gg 40 \cdot 39 * 79$ or $* \quad 69$
31. (c): The square of the number at the bottom is equal to the product of the two upper numbers*. Thus.
In fig. (A). $4 \times 9 = 6^2 = 36$
In fig. (B). $9 * 16 = 12^2 = 144$.
Let the missing number in fig. (C) be x.

Then. $16 \times * = 20^2 = 400$ or $* \ll \sim \cdot 25$.

32. (d): In fig.(A). $(6 \times 3) + (5 \times 15) = 18 + 75 = 93$.
In fig.(C). $(4 \times 8) + (18 \times 1) = 32 + 18 = 50$.
In fig. (B), missing number - $(9 \times 6) + (7 \times 5) = 54 + 35 = 89$.
33. (6): In fig. (A), $(16 - 6)^2 + (5 - 2)^2 + 10^2 + 3^2 = 109$.
In fig. (B). $(22 - 15)^2 + (21 - 19)^2 = 7^2 + 2^2 = 53$.
In fig. (C), missing number = $(17 - 13)^2 + (51 - 48)^2 - 4^2 + 3^2 = 25$.
34. (c):
$$\begin{array}{r} 2 \\ 4 \end{array}$$

In fig. (A),
$$\begin{array}{r} 3 \\ + \\ 5 \\ 4 \\ + \end{array}$$

In fig. (B), H -
$$\begin{array}{r} 8 \\ + \\ 4 \end{array}$$

Let missing number in fig. (C) be x. Then,
$$\begin{array}{r} 8 \\ + \\ 10 \\ + \\ 6 \end{array}$$

 $10 \times x - 6 = 90 \text{ or } 24x = 96 \text{ or } x = 4$.
35. (b) In the first column, $12 \times 6 = 72$; $18 + 6 - 3$.
In the third column, $16 \times 8 = 128$; $32 + 8 - 4$.
In the second column, $14 \times 8 = 112$.
So, missing number = $24 + 8 - 3$.
36. (a): Clearly, sum of numbers in each row is 17.
So, missing number = $17 - (4 + 7) = 6$.
37. (d): The number in the second column is three times the difference between the numbers in the third and first columns.
So, missing number = $3 \times (16 - 7) - 3 \times 9 = 27$.
38. (a): Clearly, $(1\text{st row})^2 + (2\text{nd row})^2 + (3\text{rd row})^2 = 4\text{th row}$.
Thus, in the first column, $4^2 + 2^2 + 1^2 = 21$.
In the second column, $5^2 + 3^2 + 8^2 = 98$.
In the third column, missing number = $6^2 + 7^2 + 3^2 = 36 + 49 + 9 = 94$.
39. (6): In the first column. $3 \times 100 + 5 \times 9 = 345$.
In the second column. $4 \times 100 + 6 \times 10 = 460$.
In the third column, missing number = $5 \times 100 + 7 \times 11 = 577$.
40. (6): Clearly, $(1\text{st row}) \times (2\text{nd row}) \times (3\text{rd row}) = 4\text{th row}$.
In the first column, $6 \times 5 \times 4 = 120$.
In the second column, $6 \times 7 \times 3 = 126$.
Let the missing number in the third column be x.
Then, $8 \times 5 \times x = 320$ or $x = 8$.
41. (a): Clearly, $(3\text{rd column})^2 + (2\text{nd column})^2 = (1\text{st column})^2$
In the first row. $5^2 + 12^2 = 13^2$.
In the second row, $8^2 + 15^2 = 17^2$.

Let the missing number in the third row be x .

Then $x^2 \cdot 24^2 \cdot 25^2$ or $x^2 + 576 = 625$ or $x^2 = 49$ or $x = 7$.

42. (6): In the first row, $6 \times | - 9, 6 \times | \gg 15$.

In the second row, $8 \times \frac{3}{-} = 12, 8 \times \frac{5}{-} = 20$.

$\therefore$ In the third row, missing number = $4 \times \wedge = 10$.

43. (rf) : Clearly. $(1\text{st row})^3 + (2\text{nd row})^3 + (3\text{rd row})^3 = 4\text{th row}$.

So. in the first column, $2^3 \cdot 1^3 + 3^3 = 8 + 1 + 27 \ll 36$.

In the third column, $0^3 + 4^3 \cdot 3^3 = 0 + 64 + 27 = 91$.

$\therefore$ In the second column, missing

number = $4^3 \cdot 2^3 + 1^3 \cdot 64 \cdot 8 \cdot 1 \cdot 73$.

44. (ib) : In the first row. $11 \times 2 \cdot | - 26$.

In the second row. $6 \times 2 + \frac{0}{-} = 16$.

In the third row. missing number = $5 \times 24 - \frac{12}{-} - 10 \cdot 6 \gg 16$.

45. (c>) : The sequence in first column is $\times 5$. Thus, $1 \times 5 = 5, 5 \times 5 = 25, 25 \times 5 = 125$.

The sequence in third column is $\times 2$. Thus, $7 \times 2 = 14, 14 \times 2 = 28, 28 \times 2 = 56$

The sequence in second column is $\times 4$.

Missing number = $12 \times 4 = 48$.

46. (id) : In the first column. $13 + 7 \ll 2 \gg 27$.

In the second column, $54 + 45 \times 2 = 144$.

Let the missing number in the third column be x .

Then, $x + 32 \cdot 2 = 68$ or $x = 68 - 64 = 4$.

47. (c) : In the first column. $29 - 8 \gg 7 \times 3 = 21$.

In the second column, $19 - 7 = 4 \times 3 = 12$.

Let the missing number in the third column be x .

Then, $31 - 6 = 5 \times x$ or $5x = 25$ or $x = 5$.

48. (6) : In the first row. $3 \times 4 \cdot 3 = 15$.

In the second row. $7 \times 5 \cdot 3 = 38$.

In the third row, missing number = $3 \times 5 + 3 = 18$.

49. (a) : In the first row, $(42 - 38) \times 11 = 44$.

In the second row. $(28 - 23) \times 11 = 55$.

In the third row. missing number = $(39 - 37) \times 11 \oplus 2 \times 11 \gg 22$.

50. (id) : In the first column. $2 \cdot 1 \cdot 1 = 3$.

In the second column. $14 \times 7 \cdot 7 = 105$.

Let the missing number in the third column be x .

Then, $x \cdot 9 \cdot 9 = 117$ or $9x = 108$ or $x = 12$.

51. (6) : In the first row, $(9 \cdot 6 \cdot 3) - (8 \cdot 4 + 4) = 2$.

* In the second row. missing number = $(4 \cdot 6 \cdot 4) - (9 \cdot 0 \cdot 3) = 14 - 12 = 2$.

52. (ic) : In the first column, $11^2 - 1^2 = 121 - 1 = 120$.

In the second column, $7^2 - 2^2 = 49 - 4 = 45$.

$\therefore$ In the third column, missing number = $5^2 - 3^2 \ll 25 - 9 = 16$.

53. (6) : In the first row. $5 \cdot 1 \times 5, 6 \times 1 = 6, 5 \cdot 6 = 11$.

In the second row, $6 \times 4 = 24, 3 \times 2 \cdot 6, 24 + 6 \cdot 30$.

In the third row, $3 \times 5 - 15$. $4 \times 3 = 12$.
Missing number » $15 \div 12 = 27$.

54. ia): In the first row, $72 \div y = 12$ or $72 = 12 \times y$.

In the second row, $96 \div y = 8$ or $96 = 8 \times y$.

Let the missing number in the third row be x .

Then, $108 \div x = 18$ or $108 = 18 \times x$ or $x = 6$.

55. id) : In the first row, $! - | \times 5 = 20$.

• " (t

In the second row,

/. In the third row, missing number =

56. (a): In the second row, $2 \times 9 + 3 \times 17 = 18 + 51 = 69$

In the third row, $2 \times 13 + 3 \times 11 = 26 + 33 = 59$

Let the missing number in the first row be x .

Then, $2x + 3 \times 13 = 49$ or $2x = 49 - 39 = 10$ or $x = 5$.

57. id): In the first row, $\frac{^4}{4} = \frac{^7}{7} - 3$.

In the second row, $\frac{^{10}}{10} = \frac{^4}{4} = 2$.

Clearly, in the third row, we have $\frac{^{61}}{61} = \frac{^{24}}{24} = 8$.

So, missing number = 83.

58. id): In the first column, $17 - 11 = 6$. $25 - 19 = 6$.

In the second column, $12 - 6 = 6$. $34 - 28 = 6$.

Let the missing number in the third column be x .

Then, $x - 8 = 19 - 11 = 8$ or $x = 27$.

59. ib): We have : $3 \times 4 = \text{number below } 4 = 12$;

$3 \times 4 \times 5 = \text{number below } 5 = 60$;

$3 + 7 \times 12 = \text{number below } 12 = 87$.

Missing number $= 3 + 7 \times 10 = 73$.

60. ib): in the first column, $3 \times 6 \times 12 = 216$.

In the second column, $2 \times 20 \times 25 = 1000$.

∴ In the third column, missing number $= 2 \times 4 \times 64 = 512$.

61. ia): The numbers in the first row form a series $1^2, 2^2, 3^2$.

So, missing number in the first row $= 4^2 = 16$.

The numbers in the second row form the series, 1, 2, 3, 4.

The numbers in the third row form the series 2, 4, 6.

So, missing number in the third row $= 6 + 2 = 8$.

62. ib): In the first row, $7 \times 3 = 21$. $9 \times 3 = 27$.

In the second row, $4 \times 9 = 36$. $2 \times 9 = 18$.

In the third row, $9 \times 6 = 54$.

Missing number $= 4 \times 6 = 24$.

63. <n) : The letters in the second and third rows are five steps ahead of those in the first and second row respectively.

So, the missing letter will be five steps ahead of F, which is IC

64. (c): The letter in the second column is three steps behind that in the first column, and the letter in the third column is four steps behind that in the second column.
So, the missing letter in the first row will be three steps behind Z, which is W.
The missing letter in the second row will be four steps behind O, which is K.
The missing letter in the third row will be three steps ahead of G. which is J.

65. (6): Putting A = 1, B = 2, Z = 26, we get the given matrix as

6	9	15
1	10	11
5	13	?

Clearly, 1st column + 2nd column = 3rd column.
Now. 5 + 13 = 18.

So, the missing letter is the 18th letter of the alphabet, which is R.

66. (a): Putting A = 1, B = 2.....Z = 26, we get the given matrix as #

8	11	17
3	7	15
5	10	?

Clearly, 2 x (2nd column - 1st column) = 3rd column - 2nd column.

Let the missing letter be the nth letter of the alphabet

Then, 2 x (10 - 5) = n - 10 or n - 10 = 10 or n = 20.

So, the missing letter is the 20th letter, which is T.

67. (c): The letters in the first row follow the sequence +5, +7.
The letters in the second row follow the sequence +6, +8.
In the third row, the first letter G moves 7 steps forward to give the second letter N.
Clearly, the missing letter will be 9 steps ahead of N i.e. W.
68. (d): Starting from M, all the consecutive letters appear in the given matrix. Also, starting from M, one can reach U, without lifting the pen even once.
69. (rf): In each column, out of the letters A, B and C, each of these must appear once. Along the diagonals, the sum of two numbers is equal to the third number.
/. The missing number will be (7 + 9) = 16 and the letter will be C.
So, the answer is 16C.
70. (a): In each row, out of the letters A, B and C, each of these must appear once. In each column, the product of the first and third numbers is equal to the second number.
So, the missing number will be (2 x 5) = 10 and the letter will be C.
Thus, the answer is 10C.
71. (6): Each row consists of alternate letters. Thus, first row contains letters V, X, Z ; third row contains letters P, R, T.
So, the missing letter is E.
The numbers in each column form an arithmetic series. Thus, first column contains numbers 4, 5, 6. The second column contains numbers 2, 3, 4.
The numbers in the third column will form the series 9, 12, 15.
So, the answer is E12.
72. (6): The letters in the first row form a series C, D, E (a series of consecutive letters).
The letters in the second row form a series I, K, M (a series of alternate letters).
Similarly, the letters in the third row will form the series D, G, J (a series in which each letter is three steps ahead of the previous one).
So, the missing letter is G.

Also, the number in the second column is equal to the product of the numbers in the first and third columns.
So. missing number is (4×7) i.e. 28.
Thus, the answer is 28G.

73. (a): The number in the third column in each row is obtained by multiplying the numbers in the first and second column in the same row. So,
In first row, $3 \times 5 = 15$ and in second row, $2 \times 4 = 8$.
Number to be filled in blank space = $4 \times 6 = 24$. Only *ia*) contains 24.
74. *id*) : The sum of numbers in each row and each column is 30.

Questions 75 to 78

Assume the pattern to be :

$$\begin{array}{ccccc} & & A & & \\ & B & & C & \\ D & & E & & F \end{array}$$

From questions 77 and 78, we have : $C \cdot D^2, B - F^2$...<i)
From questions 75 and 77, we have : $E = B - C$ (il)
From questions 75 and 78, we have : $A = B + F$...CU0

75. (c): Missing number = $\sqrt{81} = 9$. [From *ii*)]
76. *ic*): Missing number - $8^2 = 64$. [From *ii*)]
77. (6): Missing number = $225 \cdot 15 = 240$. [From (inT
78. *id*) : Missing number - $121 - 49 = 72$. [From *iii*)]

Questions 79 to 81

The pattern followed in the sample figure is :

$$8 \times 13 - 104, 8^2 = 64. 8 \cdot 13 \cdot 64 - 85.$$

79. (a>: Let the missing number be x.
Clearly. $14 + x \cdot 4 \cdot 196 = 221$ $x = 221 - 210 = 11$.
Missing number - 15^2 p- 225.
81. *ib*): Missing number = $3 \times 27 = 81$.
82. (c): Suppose X denotes the numbers in the first row and Y denotes the numbers in the second row.

Then, the "pattern is $X^2 - X = Y$.
Clearly, $3^2 - 3 = 6$; $8^2 - 8 = 64 - 8 = 56$; $10^2 - 10 = 100 - 10 = 90$;
 $2^2 - 2 = 4 - 2 = 2$; $1^2 - 1 = 1 - 1 = 0$.

Similarly. $5^2 - 5 = 20$.
So. the missing number is 5.

83. (c): The sum of numbers in the first, second, third. ... columns form the series 4, 9, 16, ...
Lt. $2^2, 3^2, 4^2, \dots$
Let the missing number be x.
Then. $2 + 10 + \dots + 25$ or $x \cdot 25 - 12 = 13$.
84. *ib*): The left hand number is obtained by adding the bottom two numbers. The top right hand number is the result of dividing the bottom two numbers.
Thus, $12 \cdot 3 = 15. 12 + 3 = 4$;
 $22 + 11 = 33, 22 \div 11 = 2$;
 $18 + 9 = 27, 18 \div 9 = 2$.
So. $32 + X = 36$ and $32 \div X = 8$ or $X = 4$.